

SONAR Slinky Showdown

Make invisible distance sensing visible with a slinky pulse.

Win condition: Most correct wave predictions and station solutions.

THE SCIENCE YOU ARE USING

Longitudinal waves. Push one end of a stretched slinky and a compression travels along it: a longitudinal wave, where the coils move back and forth along the direction of travel. Sound works the same way through air. The pulse reflects off the far end and returns.

Echo timing is ranging. If you know how fast a pulse travels and you time its round trip, you can compute distance. That is how SONAR and ultrasonic sensors measure range. You predict and interpret pulses at challenge stations.

HOW TO BUILD AND RUN IT

- 01 Send a pulse.** Stretch the slinky on the floor and push one end sharply to launch a compression pulse.
- 02 Watch the reflection.** See the pulse travel, reflect off the held end, and come back.
- 03 Predict before testing.** At each station, predict what the pulse will do, then test and check.
- 04 Time several round trips.** Send one pulse, count 4 to 6 reflections, time the whole sequence, then divide the total distance by the total time. That gives a steadier speed than timing one trip, the SONAR idea.
- 05 Solve the stations.** Work the challenge cards as a team, explaining each prediction.

Safety: Keep the slinky flat on the floor and never let go while it is stretched, or it can snap back and hurt someone. Do not overstretch it (about 3 m max). Keep fingers clear of coils. Allergy: None.

MATERIALS

Metal slinkies	4 to 6
Floor tape	shared
Station cards	set
Clipboards	6

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why): _____

The ONE variable we are testing: _____

Baseline result: _____

After redesign: _____

DEFEND YOUR DESIGN

What evidence made you change your design? _____

If you ran it again, what is the ONE thing you would change? _____

HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Model and predictions	35
Station performance	25
Explanation	20
Team communication	20
Total	100